

DroneMagNav

Magnetic-anomaly navigation board and route planner for drones

ESP32-S3 · BMI088 IMU · LIS3MDL magnetometer · BMP280 · u-blox MAX-M10S

WDMAM world anomaly map on microSD · particle-filter MagNav · MAVLink to the autopilot

Phone web app: offline world map, tap start/destination, great-circle route upload

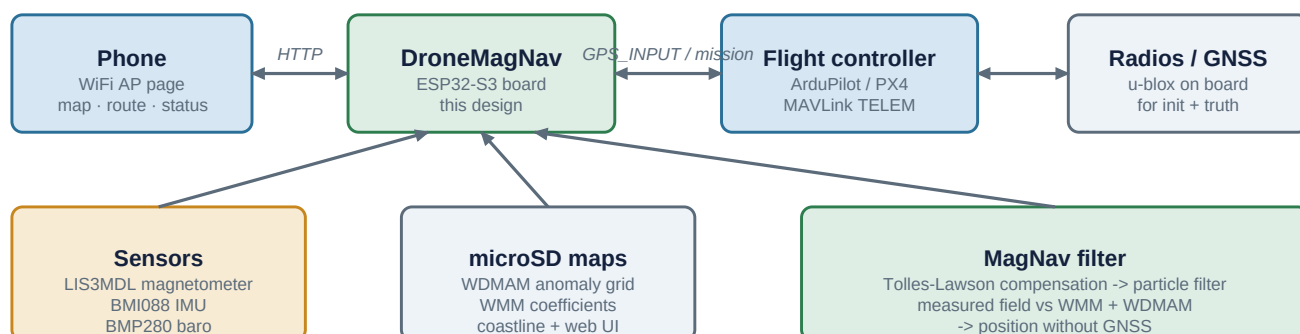
Project	DroneMagNav (GNSS-denied drone navigation)
Revision	A - schematic + routed PCB + firmware release
Date	12 September 2026
Prepared by	eng.ahmad@gmail.com · generated with KiCad 10
Design status	ERC clean · netlist verified · PCB routed, 0 electrical DRC
Classification	Prototype / experimental - not certified

1 Executive summary

DroneMagNav is a 64 x 60 mm navigation module for drones that estimates position **without GNSS** by matching the measured total magnetic field against the World Digital Magnetic Anomaly Map (WDMAM), fused with inertial dead reckoning in a particle filter. A u-blox GNSS receiver on the board initialises the filter and provides a truth reference; when GNSS is lost the MagNav estimate is streamed to the flight controller as a MAVLink *GPS_INPUT* source. A phone connects to the board's WiFi access point, shows an offline world map, and lets the operator tap a start and a destination; the board computes the shortest (great-circle) route and uploads it to the autopilot as a mission.

This release contains the complete KiCad schematic set, a routed four-layer PCB with manufacturing outputs, a bill of materials with part numbers, the ESP32-S3 firmware with the navigation algorithms and web interface, and the tools that convert the WDMAM data for the microSD card.

2 System architecture



Signal path: sensors -> compensation -> particle filter against the magnetic map -> position -> autopilot; route planned on the phone.

Figure 1 - System context

The Earth's crustal magnetic anomalies form a fixed, GNSS-independent map. Subtracting the smooth core field (World Magnetic Model) from the compensated scalar measurement leaves the anomaly, which the particle filter compares with the WDMAM grid at every particle's position. The filter is drift-free but its resolution is bounded by the map: the 3-arc-minute WDMAM grid (upward-continued to 5 km) gives kilometre-class positioning; a local aeromagnetic survey loaded in the same format brings that to tens of metres at drone altitude.

3 Hardware design

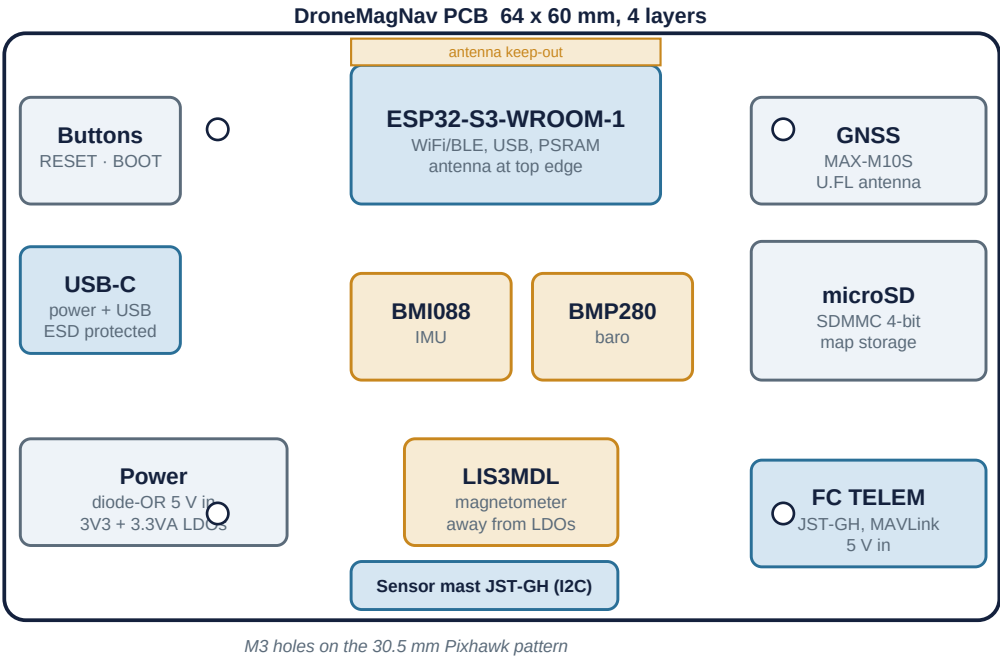


Figure 2 - Board floorplan

Block	Part	Notes
MCU / radio	ESP32-S3-WROOM-1-N16R8	16 MB flash, 8 MB PSRAM, WiFi/BLE, native USB; UART0 pads carry the GNSS
Magnetometer	LIS3MDL	16-bit, +-4 gauss, ~40 nT after averaging; SPI
IMU	BMI088	accel + gyro, drone grade; SPI, dual chip-select
Barometer	BMP280	altitude input to the field model; SPI
GNSS	u-blox MAX-M10S + U.FL	init and truth; PPS to GPIO16
Storage	microSD (Hirose DM3AT)	SDMMC 4-bit, card detect
Interfaces	USB-C, JST-GH TELEM, JST-GH mast	USBL6 ESD; Pixhawk TELEM pinout; I2C mast for RM3100
Power	SS14 diode-OR, AMS1117-3.3, AP2112K-3.3	digital 3V3 and low-noise 3.3VA sensor rail

3.1 Magnetic cleanliness

The magnetometer sits at the bottom centre, 14 mm from the regulators and away from the USB and SD connectors; the ESP32 antenna keep-out is respected in full. The dominant interference on a drone is the motors and ESC currents, which no board layout can remove: the firmware's Tolles-Lawson calibration models it, and the sensor-mast connector J5 allows an external RM3100 on a boom for survey-grade use.

4 Navigation algorithms

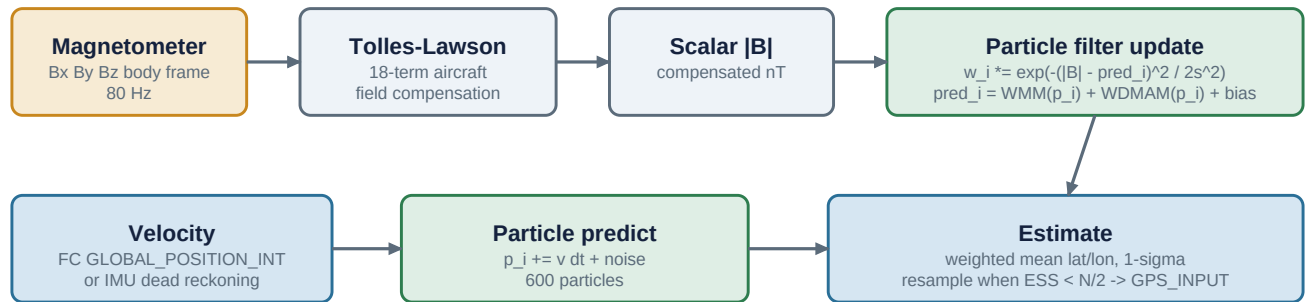


Figure 3 - MagNav particle filter dataflow

- **WMM core field:** degree-12 spherical-harmonic model evaluated per particle; coefficients read from the NOAA WMM.COF file on the card.
- **WDMAM map window:** 96 x 96 cells of the int16 nT world grid cached in PSRAM around the estimate, bilinear interpolation, seamless longitude wrap.
- **Tolles-Lawson:** 18 terms (permanent, induced, eddy) fitted by ridge-regularised least squares from a 3-minute calibration manoeuvre with GNSS; stored on the card.
- **Particle filter:** 600 particles (lat, lon, scalar bias); velocity from the flight controller or IMU; Gaussian likelihood with 40 nT sigma; systematic resampling with roughening.
- **Routing:** haversine distance, initial bearing, great-circle intermediate points at the chosen spacing; uploaded as NAV_TAKEOFF + NAV_WAYPOINT items.

5 PCB implementation

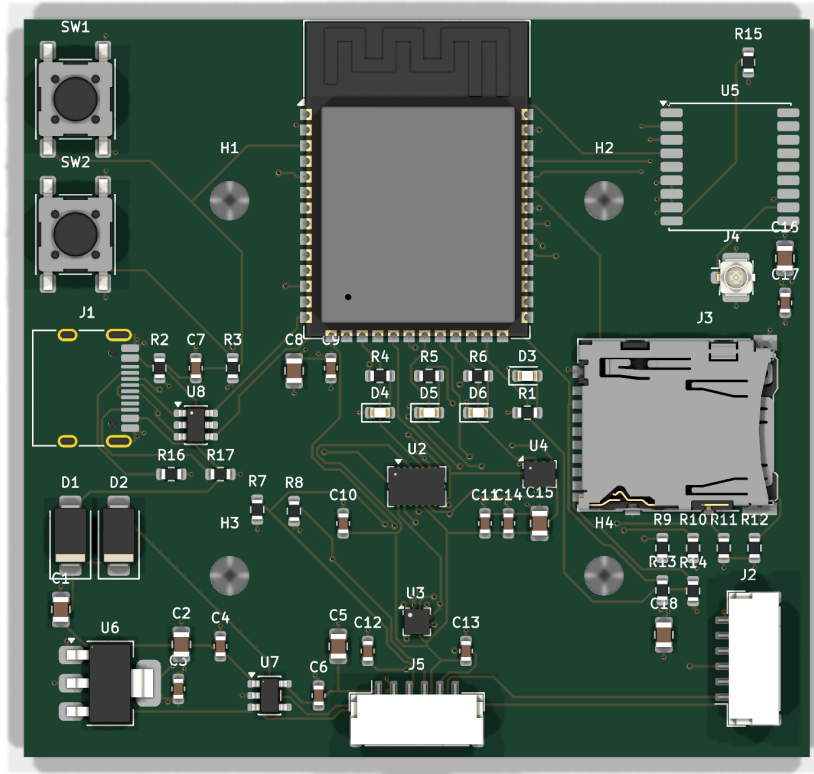


Figure 4 - Routed board, top side

Four layers: signals on the outer layers, solid GND (In1) and +3V3 (In2) planes inside, both notched under the module antenna. Autorouted with Freerouting constrained to the outer layers and the keep-outs, then scripted power fanout to the planes and hand-verified routes for the GNSS UART and the I2C bus. Rules: 0.2 mm track, 0.15 mm clearance, 0.45/0.2 mm vias. Power vias are dog-boned outside their pads (74 moved by tools/dogbone_vias.py); only three 0.4 mm vias remain in pads where the 0.5 mm-pitch BMI088 escapes and the microSD routing leave no room (U2 pads 2/4, J3 pad 4) and the fabrication notes require them to be filled and capped. Fab outputs: Gerbers, Excellon drill + map, pick-and-place, FABRICATION-NOTES.md, Gerber ZIP.

6 Bill of materials

References	Qty	Value	Manufacturer P/N
C1, C2, C5, C8, C15, C16, C18	7	10uF	CL21 series
C3, C6, C9, C10, C11, C12, ...	9	100nF	CL10 series
C4, C7	2	1uF	CL10 series
D1, D2	2	SS14	SS14
D3	1	PWR	19-217/GHC-YR1S2/3T
D4	1	NAV	19-217/GHC-YR1S2/3T
D5	1	FIX	19-217/BHC-ZL1M2RY/3T
D6	1	LINK	19-217/Y2C-CQ2R2L/3T
J1	1	USB-C	TYPE-C-31-M-12

References	Qty	Value	Manufacturer P/N
J2	1	FC_TELEM	SM06B-GHS-TB
J3	1	microSD	DM3AT-SF-PEJM5
J4	1	GNSS_ANT	U.FL-R-SMT-1(10)
J5	1	SENSOR_MAST	SM06B-GHS-TB
R1	1	1K	RC0603FR-071KL
R2, R3, R9, R10, R11, R12, ...	9	10K	RC0603FR-0710KL
R4, R5, R6	3	330R	RC0603FR-07330RL
R7, R8	2	4.7K	RC0603FR-074.7KL
R16, R17	2	5.1K	RC0603FR-075.1KL
SW1	1	RESET	PTS645SM43SMTR92
SW2	1	BOOT	PTS645SM43SMTR92
U1	1	ESP32-S3-WROOM -1-N16R8	ESP32-S3-WROOM-1-N16R8
U2	1	BMI088	BMI088
U3	1	LIS3MDL	LIS3MDLTR
U4	1	BMP280	BMP280
U5	1	MAX-M10S	MAX-M10S-00B
U6	1	AMS1117-3.3	AMS1117-3.3
U7	1	AP2112K-3.3	AP2112K-3.3TRG1
U8	1	USBL6-2SC6	USBL6-2SC6

7 Firmware and phone interface

PlatformIO / Arduino-ESP32 project (firmware/): register-level SPI drivers for the three sensors, NMEA parser for the GNSS, a minimal MAVLink v2 stack (heartbeat, GPS_INPUT, mission upload, position/velocity intake), the navigation modules above, and a WiFi access point serving the phone page from the card. Flashing is over USB-C (native USB bootloader, no programmer). The phone page draws the Natural Earth coastline offline, pans/zooms by touch, places START/DEST by tap, shows the great-circle line, distance and bearing, and uploads the mission.

HTTP endpoint	Purpose
GET /api/status	mode, position, sigma, field, anomaly, heading, link states
GET /api/route?lat1&lon1;&lat2;&lon2;&alt;&spacing;	great-circle waypoints, distance, bearing
POST /api/upload	send the last route to the flight controller as a mission
GET /api/anom?lat&lon;&span;&n;	anomaly grid for the heat overlay
POST /api/cal?start=1 0	Tolles-Lawson calibration start/stop
POST /api/init?lat&lon;	force MagNav mode from a known position

8 Verification

Check	Tool	Result
Electrical rule check (5 sheets)	kicad-cli sch erc	0 errors, 1 warning (BMI088 SDO1/SDO2 on MISO, per Bosch)
Netlist connectivity, 21 critical nets	tools/check_netlist.py	21 / 21 pass
PCB connectivity	kicad-cli pcb drc	0 unconnected pads
PCB electrical rules	kicad-cli pcb drc	0 violations (22 silkscreen notes)
WMM core-field evaluator vs NOAA WMM2025 test values	tools/wmm_check.py	100 / 100 points, worst error 0.001 nT
WDMAM SD-card grid vs source file (7200 x 3601 cells, 51.9 MB)	tools/wdmam_check.py	225 sample cells identical, 100 % coverage
MAX-M10S straps vs u-blox integration manual UBX-20053088 R05	manual, table 1 / sec. 4.1	VIO_SEL open = 3.3 V I/O (as built); TIMEPULSE-SAFEBOOT_N tie honoured in firmware
Vias in pads after dog-bone pass	tools/fab_audit.py	3 in small pads (listed for fill & cap), 2 in thermal pads, 0 unconnected, 0 DRC
Firmware build (PlatformIO, Arduino-ESP32 3.3.11)	pio run	SUCCESS, 0 warnings; flash 1.08 MB (25.7 %), RAM 94 KB (29.3 %); images in firmware/release

9 Limitations and next steps

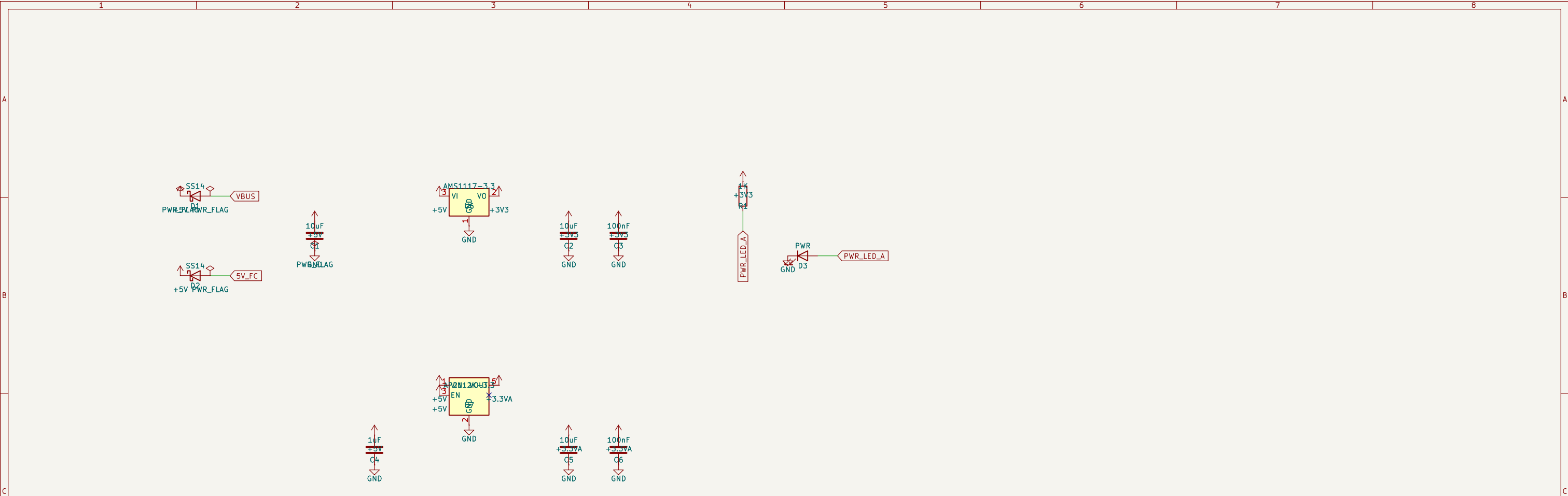
- WDMAM resolution and 5 km altitude limit stand-alone accuracy to kilometre class; add a local survey map for metres.
- LIS3MDL is adequate for the world grid; use an RM3100 on the mast connector for survey-grade work.
- Order the board with filled-and-capped vias (three vias-in-pad, see FABRICATION-NOTES.md in the Gerber ZIP).

- Flash firmware/release/dronemagnav-firmware.factory.bin, run the calibration flight, benchmark MagNav against GNSS before relying on it.

Appendix A - Schematic sheets

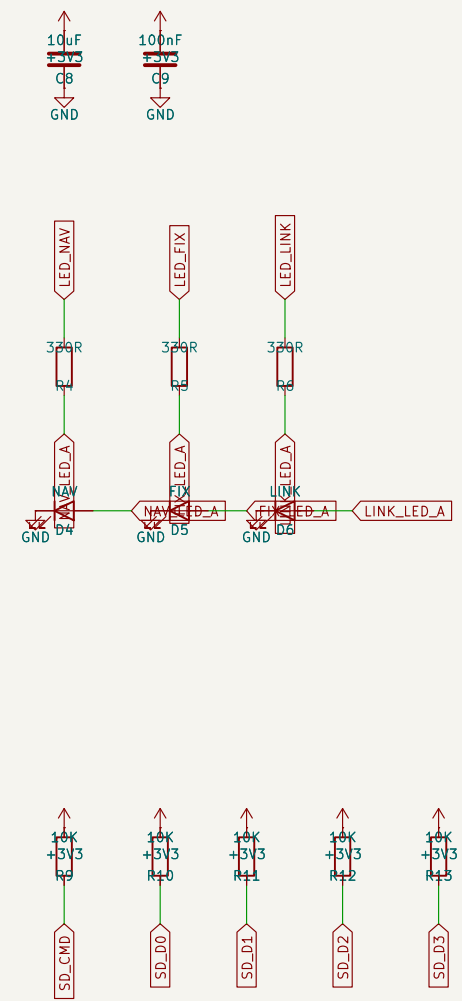
Root, power, MCU, sensors, GNSS + storage, interfaces - as plotted from KiCad. Every pin carries a drawn wire stub with its net label; the netlist is the authoritative artifact validated in Section 8.



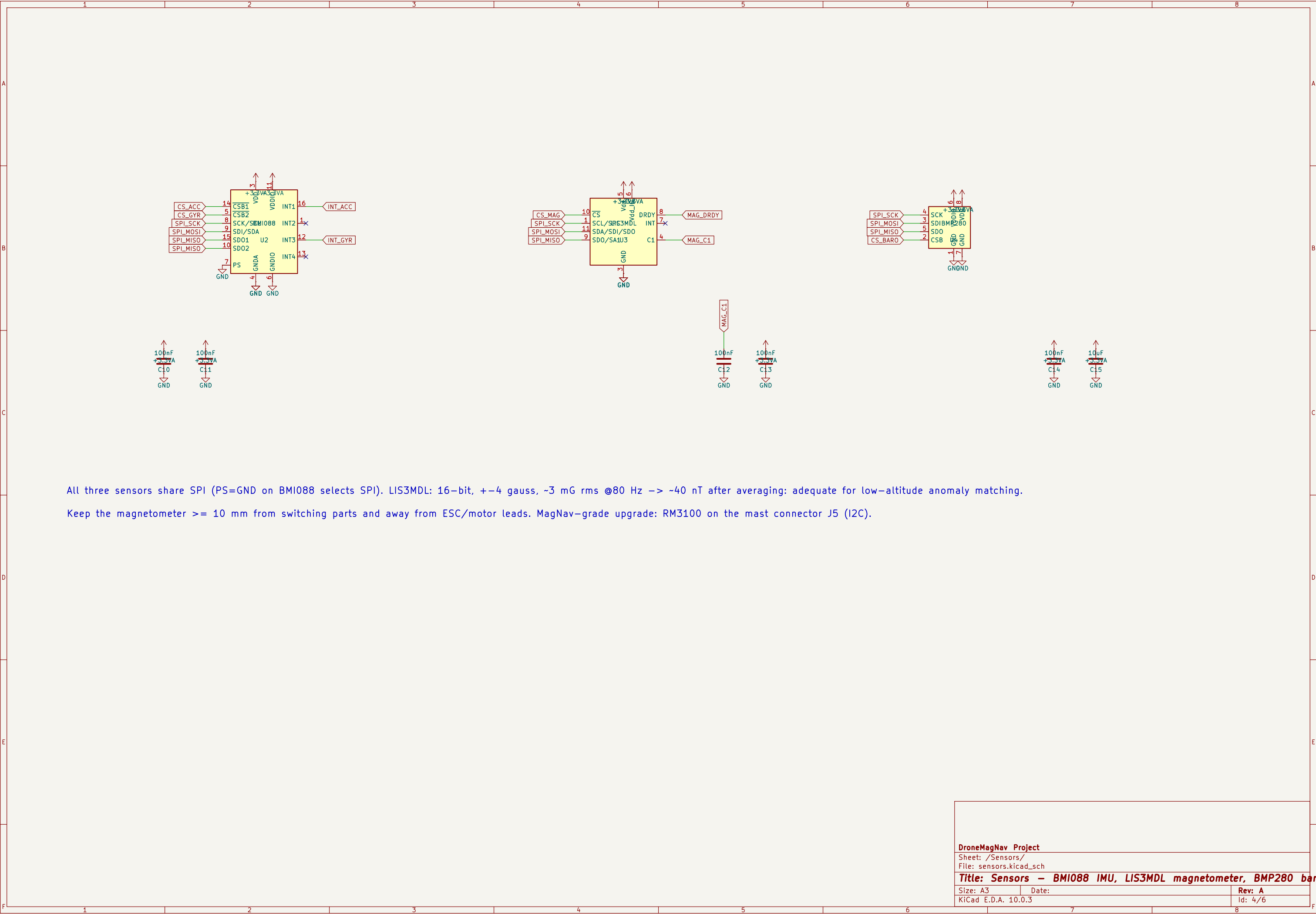


Two 5 V sources diode-ORed: USB-C VBUS (bench) and the flight-controller TELEM port (5V_FC, in flight). SS14 drop ~0.35 V.

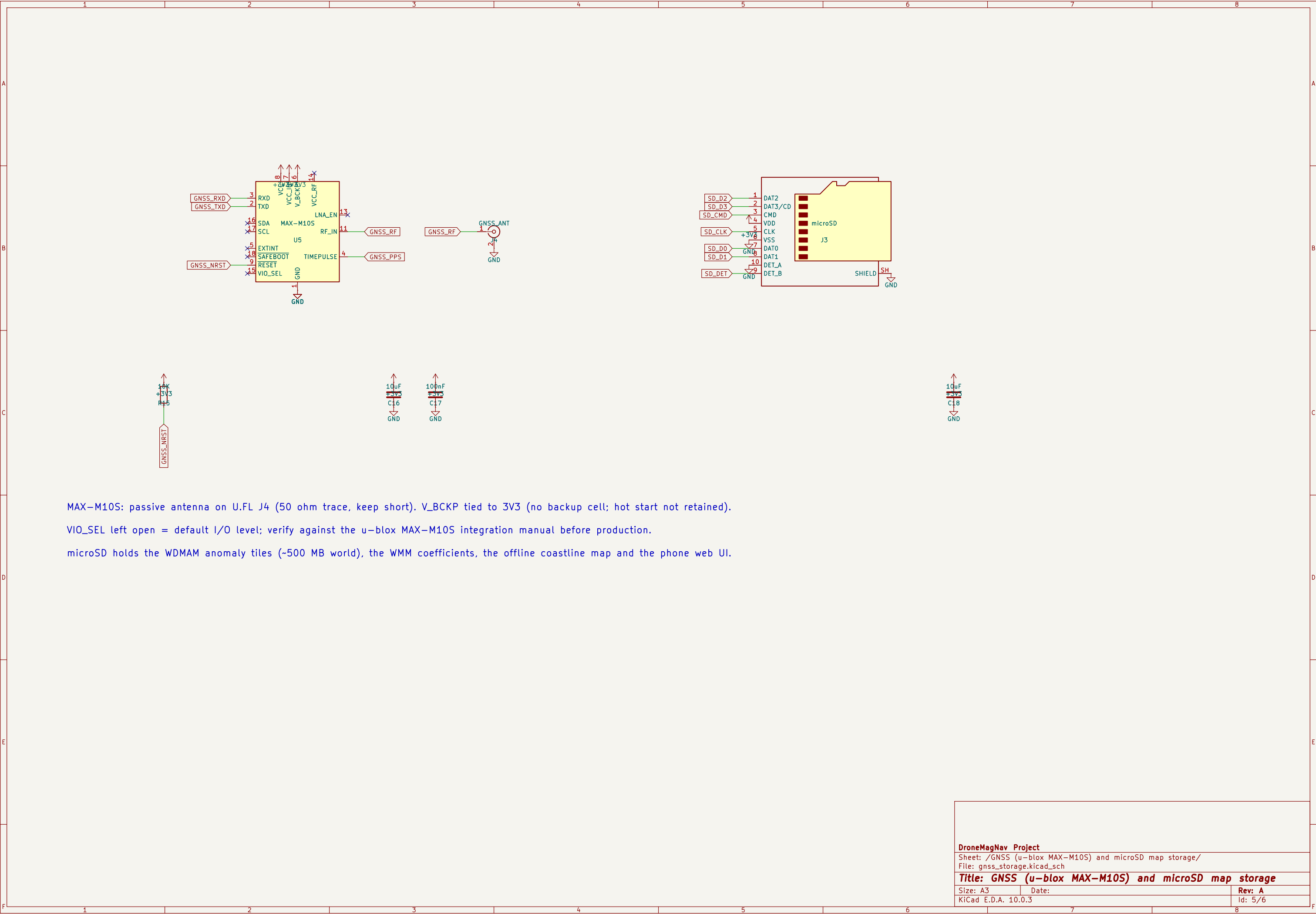
AMS1117 (1 A) feeds ESP32/SD/GNSS; AP2112K (low-noise LDO) feeds the IMU/magnetometer/baro on the +3V3A rail.

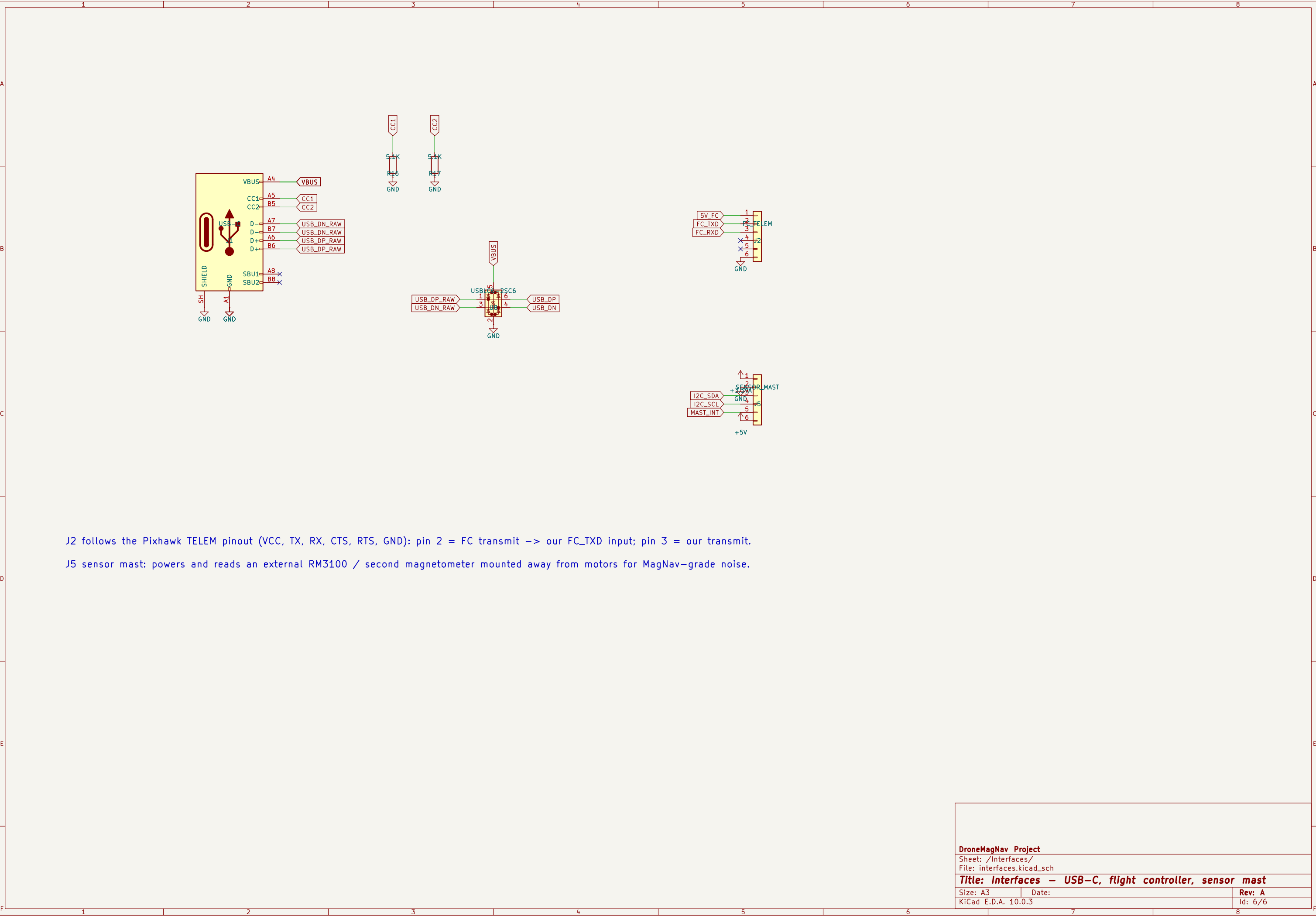


SPI @10 MHz to IMU/mag/baro; SDMMC 4-bit to microSD; UART0 pads (GPIO43/44) GNSS; UART2 MAVLink to flight controller; I2C to external sensor mast.



All three sensors share SPI (PS=GND on BMI088 selects SPI). LIS3MDL: 16-bit, +-4 gauss, ~3 mG rms @80 Hz -> ~40 nT after averaging: adequate for low-altitude anomaly matching.
Keep the magnetometer >= 10 mm from switching parts and away from ESC/motor leads. MagNav-grade upgrade: RM3100 on the mast connector J5 (I2C).





J2 follows the Pixhawk TELEM pinout (VCC, TX, RX, CTS, RTS, GND): pin 2 = FC transmit -> our FC_TXD input; pin 3 = our transmit.
J5 sensor mast: powers and reads an external RM3100 / second magnetometer mounted away from motors for MagNav-grade noise.